

MMT-03 Summary Document

Overview

This document summarizes the lecture MMT-03 on Algorithmic Trading Process.

The lecture covers the following topics:

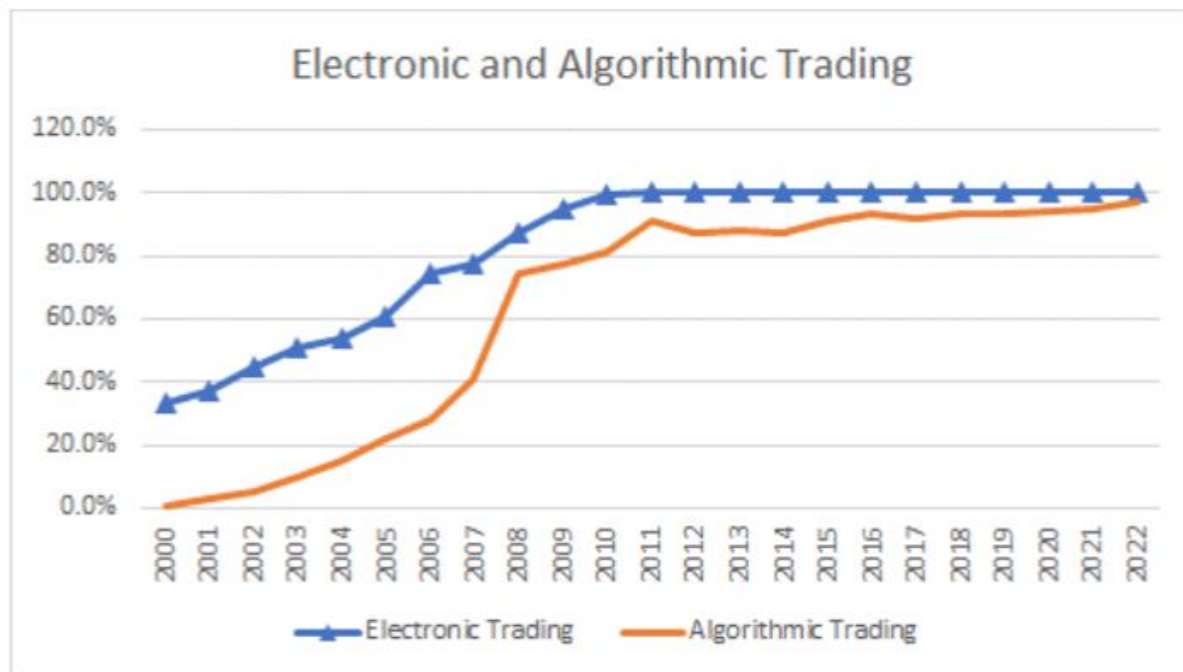
- Introduction to Electronic and Algorithmic Trading
- Market Impact, Financial Markets Landscape and Market Participants
- Trading algorithms
- Algorithmic Decision-Making Process
- Transaction Costs

Introduction to Electronic and Algorithmic Trading

Electronic Trading: It's the process of transacting orders over a computer network.

Algorithmic Trading: It's the computerized execution of financial instruments following a set of pre-specified trading rules and instructions.

Electronic and algorithmic trading has increased in market volume participation since 2000.



Institutional trading orders are usually large, e.g., much more than 100 shares. These large orders surely cause a price impact on the stock they're transacting. Institutions combat this phenomenon

by slicing their order and trade over time and balancing their market impact (MI) and timing risk (TR).

Algorithms order slicing decision is based on:

- 1) Time-Periods
- 2) Volumes
- 3) Prices

Market Impact, Financial Markets Landscape and Market Participants

Market Impact

It represents the variation in the stock price caused by a trade or an order. It's one of the costliest transaction cost components and it always results in adverse price variation for the trader impacting negatively on his performance. It can be divided into Temporary MI and Permanent MI.

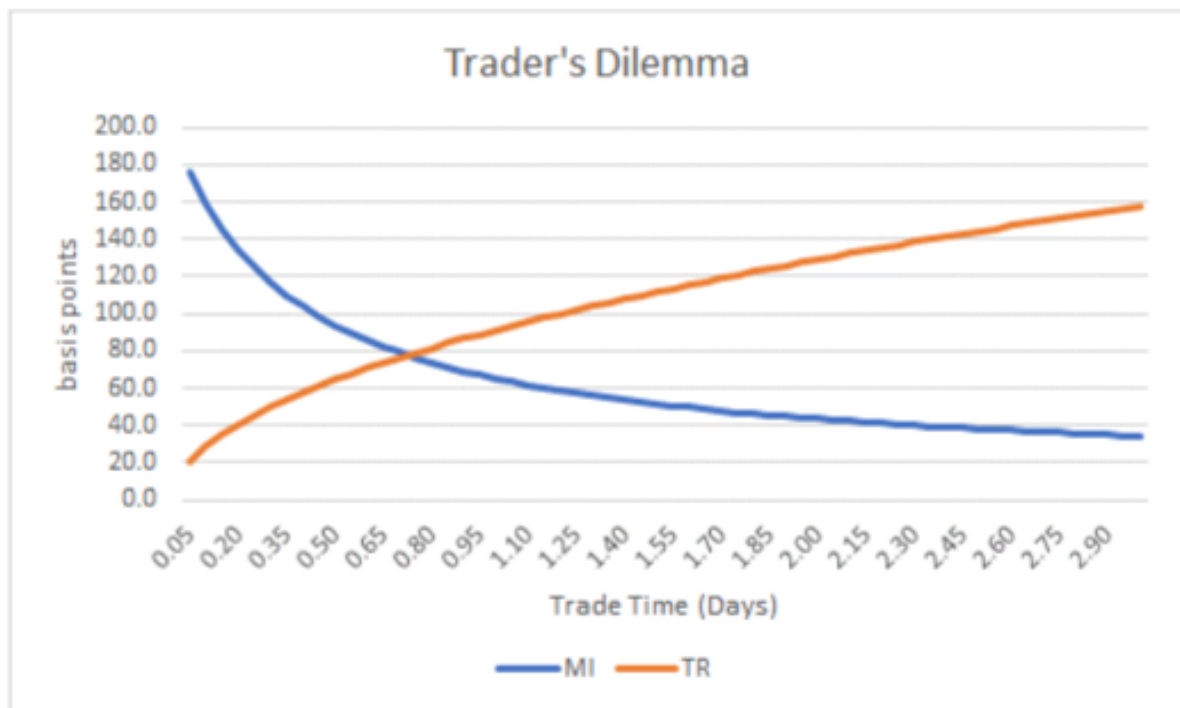


Timing risk is related to the uncertainty surrounding the estimated transaction cost. It consists of three components:

- Price volatility: This causes the stock price to be different than the estimated value.
- Liquidity risk: Share volume, aggregate buying and selling pressure drive market impact cost
- Parameter estimation error: The standard error surrounding the market impact model parameters.

Market impact is a decreasing function with trade time. Trading faster incurs higher MI and trading slower incurs less MI.

Timing Risk is an increasing function with trade time. Trading faster incurs lower TR and trading slower incurs higher TR.



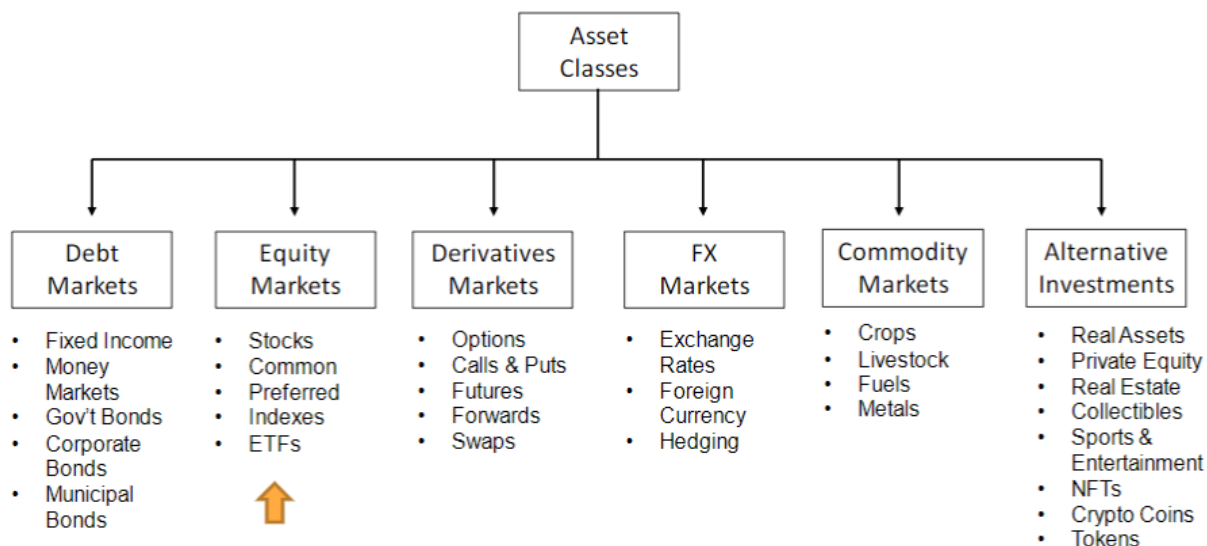
Trader's Dilemma: Trading too fast will incur too much market impact cost but trading too slow will incur the order too much market exposure risk. In other words, trade too fast and you move the market but trade too slow and the market moves you.

Financial Markets Landscape

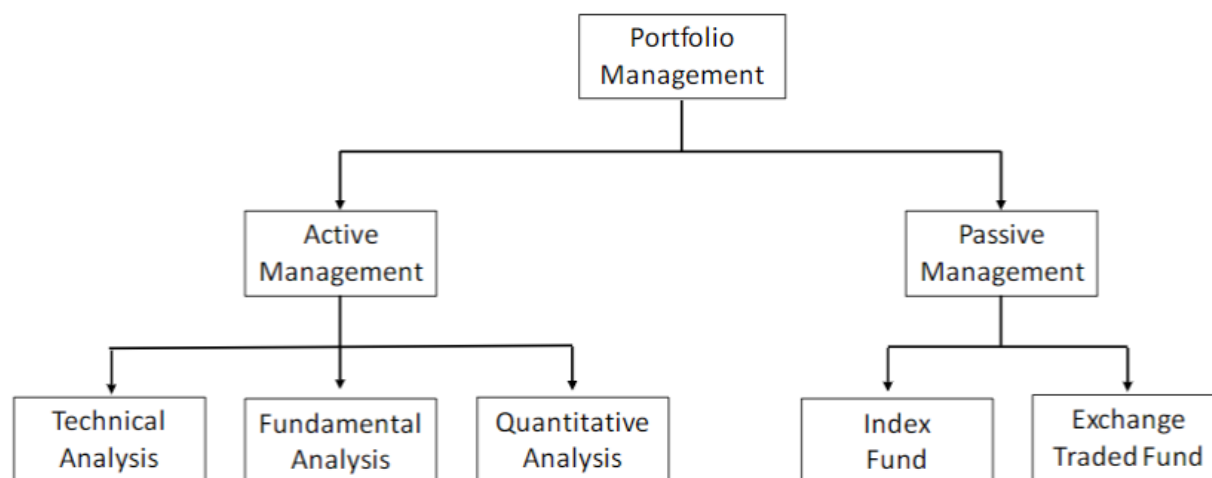
The investment cycle consists of 4 steps:

- Asset Allocation
- Portfolio Construction
- Implementation
- Portfolio Attribution

Asset classes can be divided into



Portfolio Management can be divided into



As traders, we have different motivations while making investment decisions:

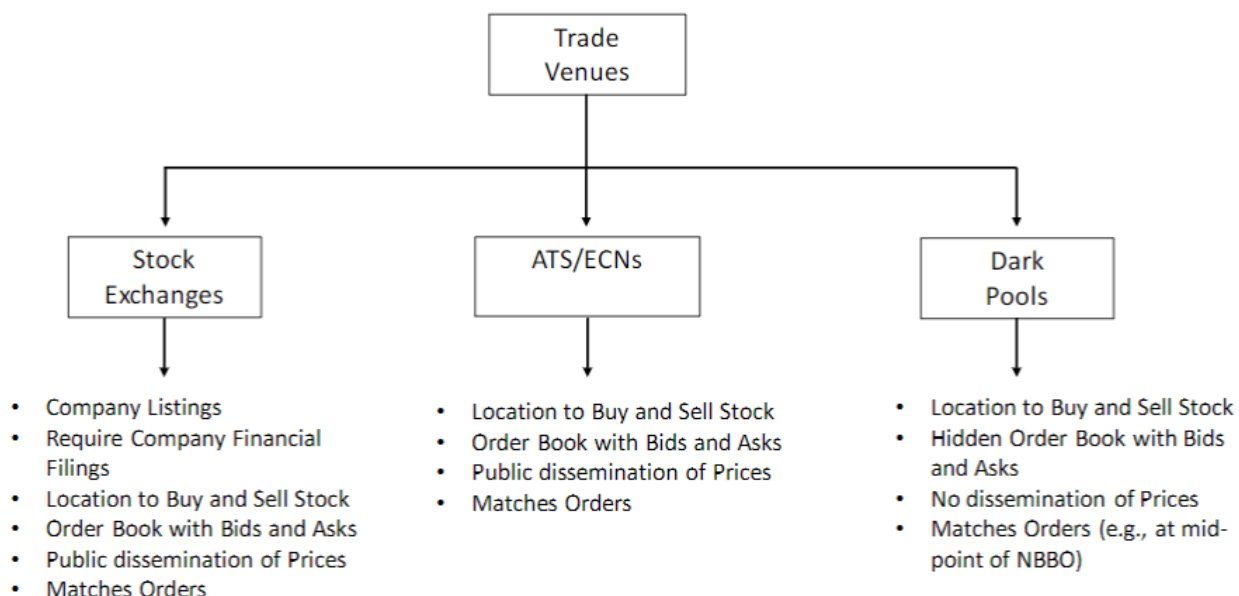
- Cash Inflow
- Cash Redemption
- Income/Dividend Payment
- Short-Term Alpha
- Long-Term Alpha
- Stock Mispricing
- Company Fundamentals
- Quantitative Models
 - Portfolio Optimization
 - Portfolio Rebalance
- Index Change / Reconstitution
- Risk Management
 - Risk Constraints
- Quant Signals
 - Price Target
 - Market Signal
- Statistical Arbitrage
 - Index/ETF-Arb
 - Stat-Arb
 - Risk-Arb/Merger-Arb
 - Pairs-Trade

Trading desks will have different motivations and types of decisions:

Function	Cash Desk	Program Desk	Electronic Desk
Trading:	Single Stock / Blocks	Programs / Baskets Index / ETFs	Single Stocks / Baskets
Trading Concerns:	Price Movement Alpha	Risk Management Cash Balancing	Control / Anonymity Liquidity
Trade Execution:	Broker Algorithms	Broker Algorithms	Broker Algorithms Investor Algorithms (DMA)
Decision Maker:	Cash Trader (w/Investor Instructions)	Program Trader (w/Investor Instructions)	Investor (Client)

Retail investors can be classified into Cash desks.

Traders can invest in different venues as



Market Participants

There are many, such as

- Brokers: Investment banks, full service, bulge bracket, research brokers, etc
- Institutions: Asset management, mutual funds, index funds, insurance companies and so on
- Hedge Funds: Quant & non-quant
- Liquidity Providers: Specialist / market makers / AMM, DMM (NYSE), QMM (Nasdaq)
- Retail Investors: Individuals, wealth managers, millennials, Gen-Z
- Proprietary Trading Firms (PTF)
- High-Frequency Trader (HFT)

Trading algorithms

There are different types of Trading Algorithms:

- Execution-Only: Algorithms tasked with transacting an order that was entered by a portfolio manager and determined outside of the system.
- Quantitative (Profit Seeking): Algorithms that is responsible for both investment decision and the implementation of shares. The investment decision is often based on a quantitative approach such as portfolio optimization, market signal, or statistical arbitrage opportunity. These algorithms seek to profit from their investments and are used by quantitative traders and proprietary trading firms.
- Auto-Market Making (AMM): Algorithms tasked with performing the traditional specialist and market-making function of providing liquidity in the stock. E.g., NYSE = DMM and Nasdaq = QMM. AMM algorithms seek to profit off the bid-ask spread and from rebate incentives provided by the exchange for providing liquidity.

- Liquidity Trading: Algorithms that seek to profit from the bid-ask spread and from market rebates. Liquidity trading algorithms also seek to uncover buying and/or selling patterns in the stock and use that information to their advantage. Liquidity trading algorithms are often customized and used by High-Frequency Traders (HFT).
- Direct Market Access (DMA): Brokers provide connectivity and access to the various trading venues to clients. Clients employ their own algorithms for transactions but use the brokers' connectivity to the different liquidity pools.

We can also classify algorithms by

- Aggressive: Get me done. Sweep all liquidity available at my price or better. Uses more market orders during times of favourable prices to capture these better prices.
- Working Orders: Balance the tradeoff between cost and risk, manage the placement of the market and limit orders. VWAP/TWAP, POV, Implementation Shortfall, Inline.
- Passive: Usage of crossing systems and dark pools. Liquidity-seeking algorithms. It uses more limit orders than market orders.

We can slice strategies orders based on

- Time-Based: Trade a specified number of shares at specified times during the day. E.g., execute 10% of the order between 9:30 am and 10:00 pm. The same number of shares will occur in these time periods regardless of market volumes and market prices.
- Volume-Based: Participate with a specified amount of volume. E.g., POV=10% indicates that your trades will constitute 10% of market volume. More trading will occur in times of higher market volume and less trading will occur in times of lower market volume.
- Price-Based: Traders may execute more or less aggressively based on market prices.

Should we update an algorithmic trading system order? If not, we have static algorithms. If yes, our algorithm will be classified as dynamic.

Algorithm Strategies can be classified as

- VWAP / TWAP
- POV / Volume
- Arrival Price / Implementation Shortfall (IS)
- Liquidity Seeking / Dark Pool Algorithms
- Front-Load / Back Load
- MOO / MOC
- Optimal Strategy
- Portfolio / Basket Algorithms
- Dynamic

Algorithmic Decision-Making Process

Since there might be MI and TI in your trading orders, you need to be proactive in your trading process. They need to specify their trading goals. These are quantified in terms of macro and micro-level decisions and based on the investment objective of the fund.

A goal of algorithmic trading is to reduce the overall transaction costs and to ensure that the trading goal is consistent with the investment objective of the fund.

In order to implement an algorithmic trading strategy or process, we need to know the precise reasons for investment and the risk aversion of the firm. The process consists of

1. Select a broker
2. Select an algorithm and its parameters
3. Transact using LOM and SOR
4. Revise the strategy if necessary
5. Measure cost and evaluate the performance

We, as investors or traders, need to take into account the following factors:

- Investment objective: the motivation behind the trade.
- Investor strategy selection: how the investor wants to trade the order.
 - Macro-Level strategy: how the order is to be sliced into smaller pieces and traded over time based on expected market conditions.
 - Micro-Level strategies: how to determine when it is appropriate for an algorithm to deviate from an optimally prescribed strategy (based on current market conditions).
- Order submission rules: how and where an order should be entered into the market.
 - Limit Order Model (LOM): determine the best combination of limit and market orders to best adhere to the underlying execution strategy (macro and micro goal).
 - Smart Order Router (SOR): computes in real-time the likelihood of transacting at the specified prices (determined from the limit order model) at the different venues.

Trading Venues

They can be

- Listed: A stock listed on the NYSE or AMEX exchanges would trade on these exchanges via a single specialist.
- OTC (Over The Counter), e.g. as the Nasdaq OTC market via an electronic system of one or multiple market makers with their respective order books.
- Exchange: Where companies list their stock for execution. Here, the specialist is replaced by a Designated Market Maker (DMM).
- Venue (ECN/ATS): A destination where stocks trade like an exchange, however, these venues do not list companies or provide information about the company.
- Dark Pool: A crossing network is a trading destination that does not publicly display any of its orders or quotes to the public.

Venue Revenue Models can be classified as

- Maker-Taker Model
- Inverted Maker-Taker Model
- Commission Based

Revenue Model	Maker (Posting a Limit Order)	Taker (Trading with Market Order)
Maker-Taker	Receives Rebate	Pays Fee
Inverted Maker-Taker	Pays Fee	Received Rebate
Commission	Pays Fee	Pays Fee

Execution Style

There are two ways that parties trade:

- Agency Transaction: The investor pays the broker a commission to execute the order on a best-efforts basis.
- Principal Bid / Capital Commitment: The investor pays the broker a premium to execute the order.

Transaction Costs

Classified as

- Economic: dollars paid by buyers but not received by sellers and dollars paid by sellers but not received by buyers.
- Financial: dollars amount paid above the decision price (premium) for buy orders and dollars amount paid below the decision price (discount) for sell orders.

Transaction costs are an unavoidable cost of business. They consist of nine cost components:

- Commissions
- Spreads
- Taxes
- Fees & rebates
- Delay cost
- Price appreciation
- Market impact
- Timing risk
- Opportunity cost